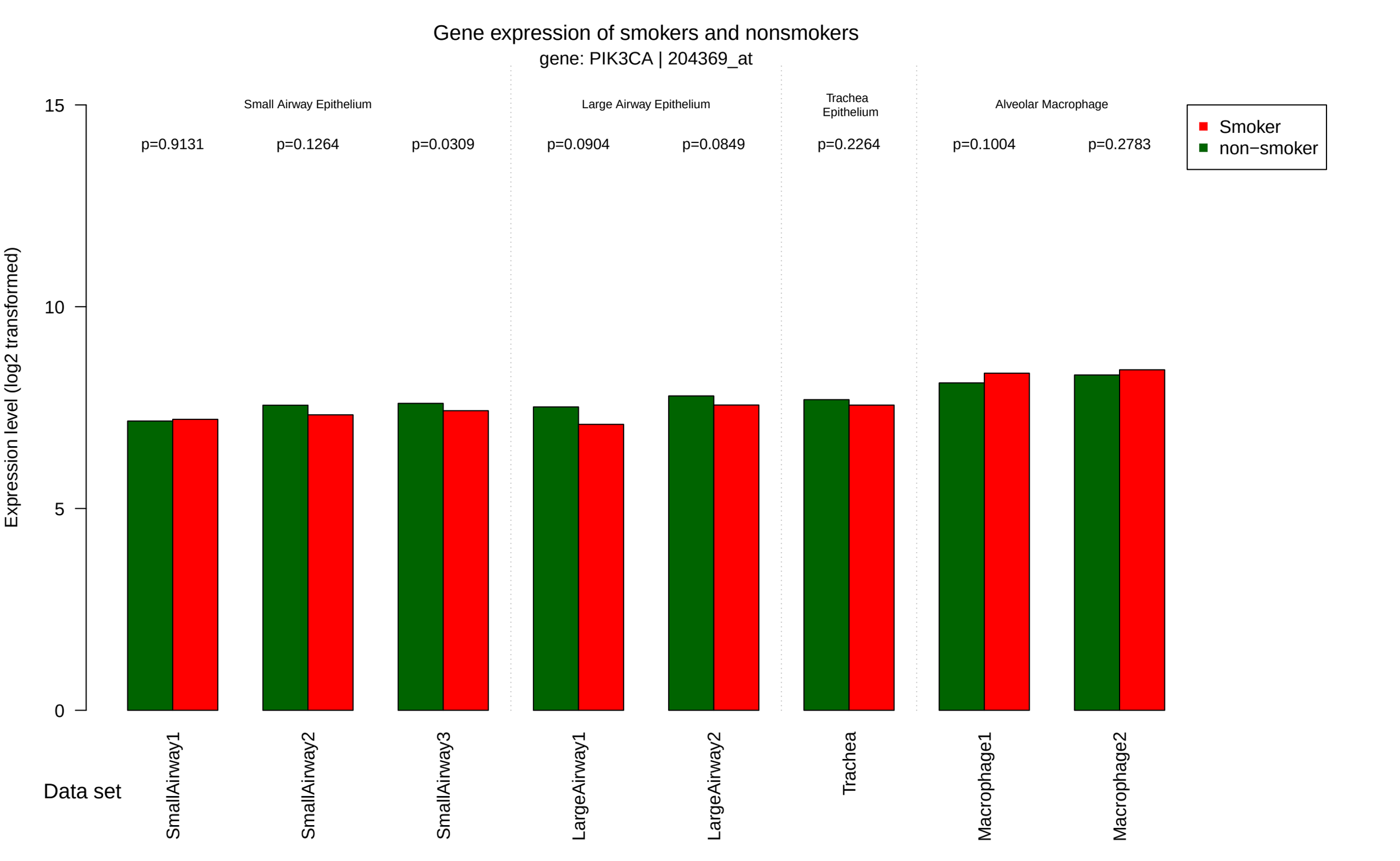
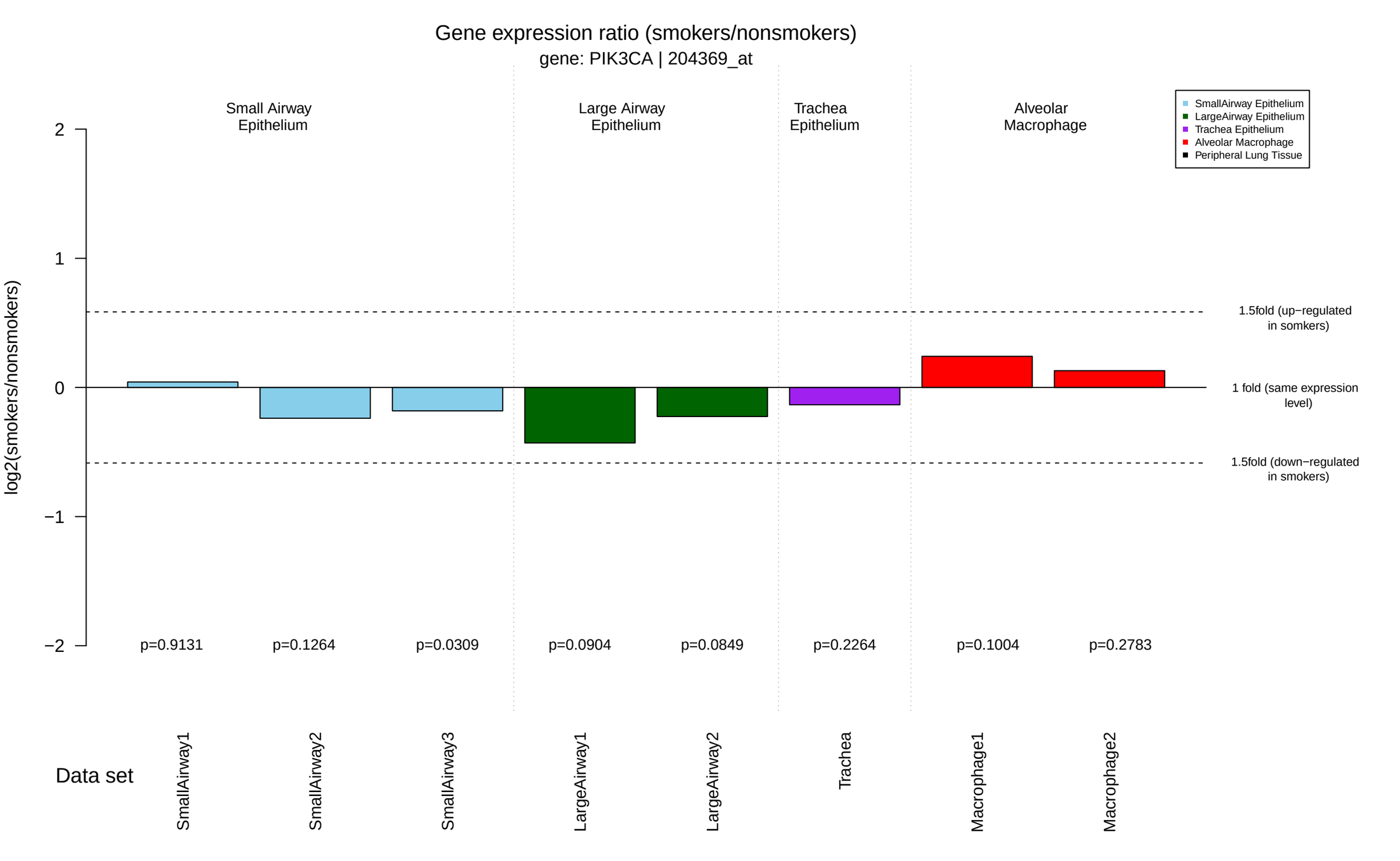
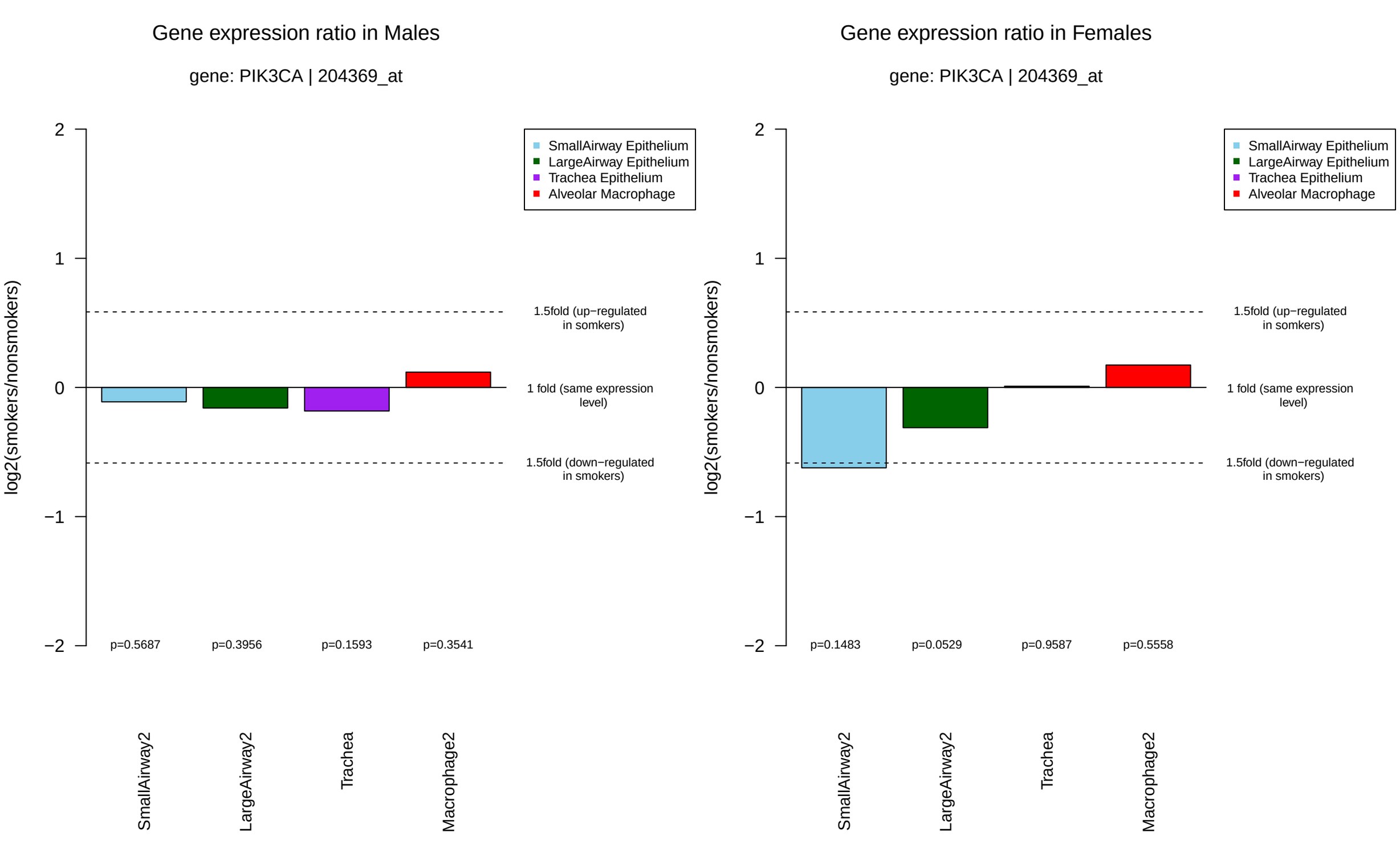
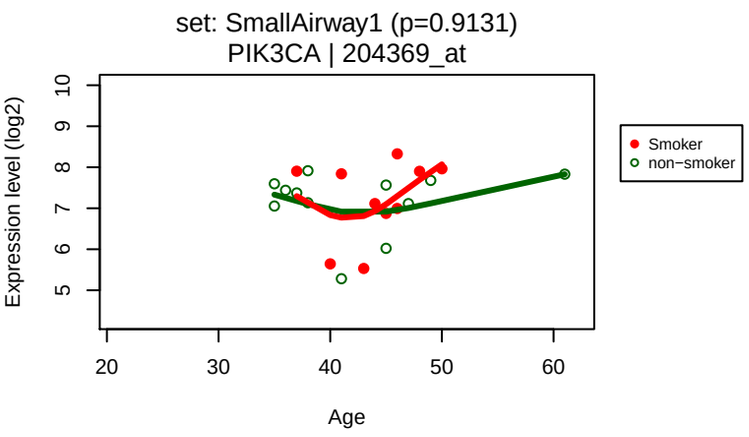
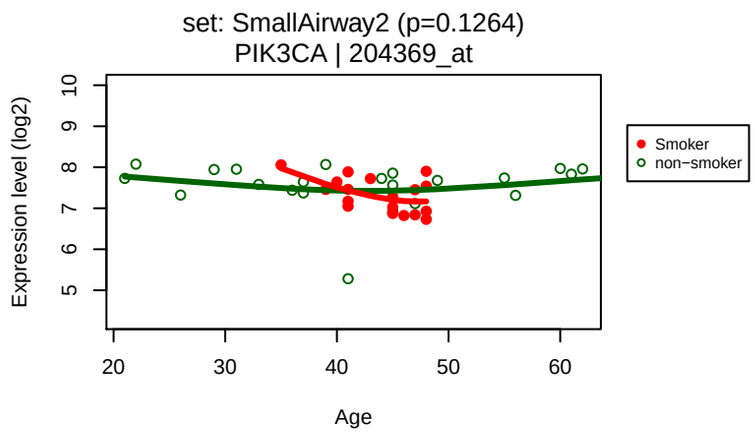
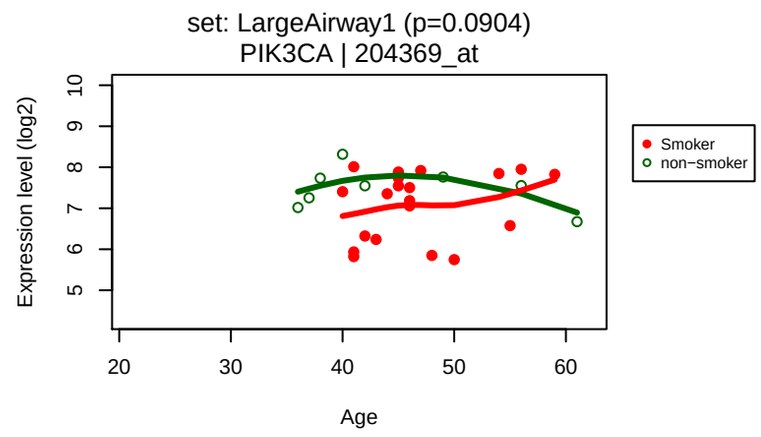

Gene expression of smokers and nonsmokers
gene: PIK3CA | 204369_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9131
p=0.1264
p=0.0309
p=0.0904
p=0.0849
p=0.2264
p=0.1004
p=0.2783
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

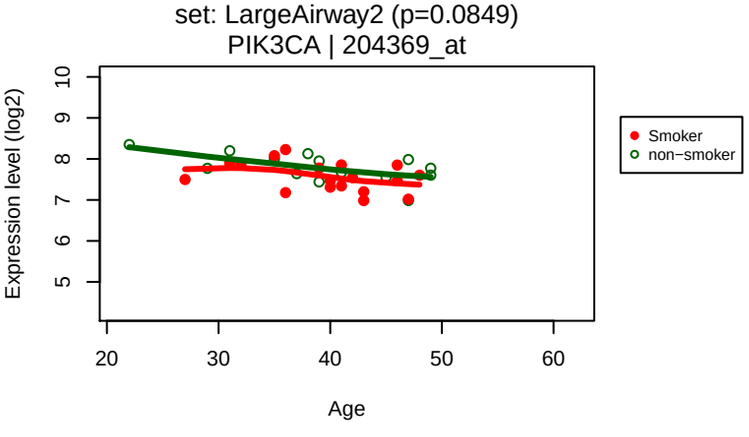

Gene expression ratio (smokers/nonsmokers)
gene: PIK3CA | 204369_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9131
p=0.1264
p=0.0309
p=0.0904
p=0.0849
p=0.2264
p=0.1004
p=0.2783
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

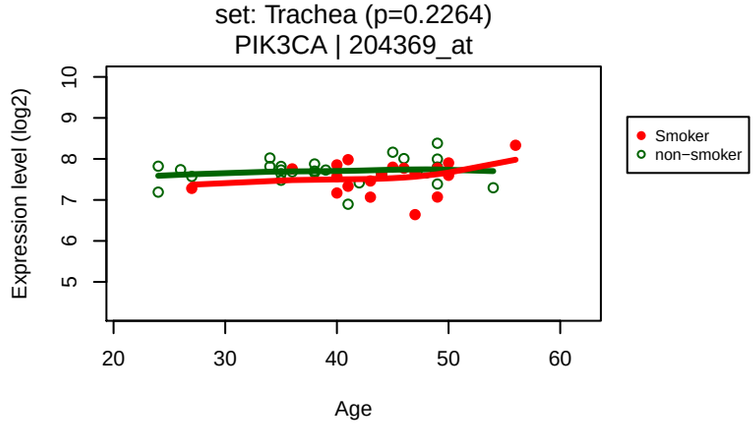

Gene expression ratio in Males
gene: PIK3CA | 204369_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5687
p=0.3956
p=0.1593
p=0.3541
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIK3CA | 204369_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1483
p=0.0529
p=0.9587
p=0.5558
SmallAirway2
LargeAirway2
Trachea
Macrophage2

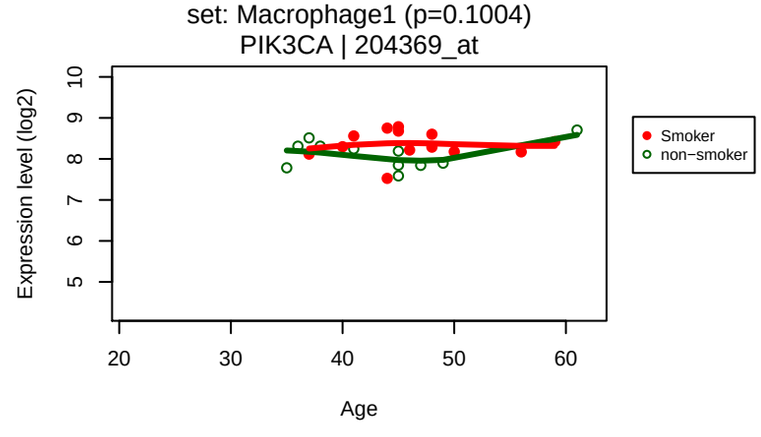

set: SmallAirway1 (p=0.9131)
PIK3CA | 204369_at
Expression level (log2)
Age
Smoker
non−smoker

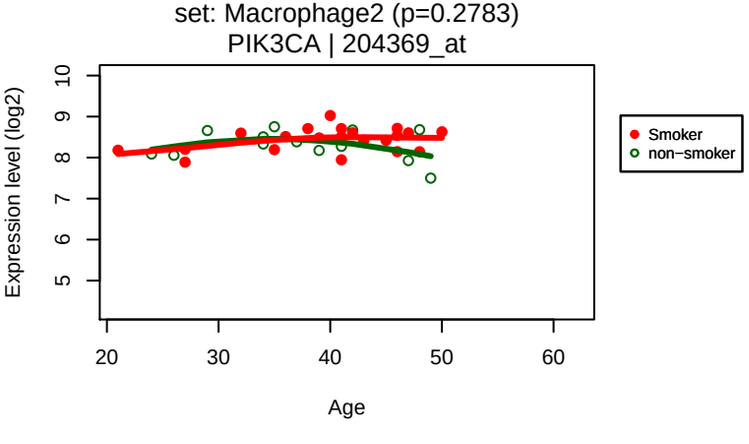

set: SmallAirway2 (p=0.1264)
PIK3CA | 204369_at
Expression level (log2)
Age
Smoker
non−smoker

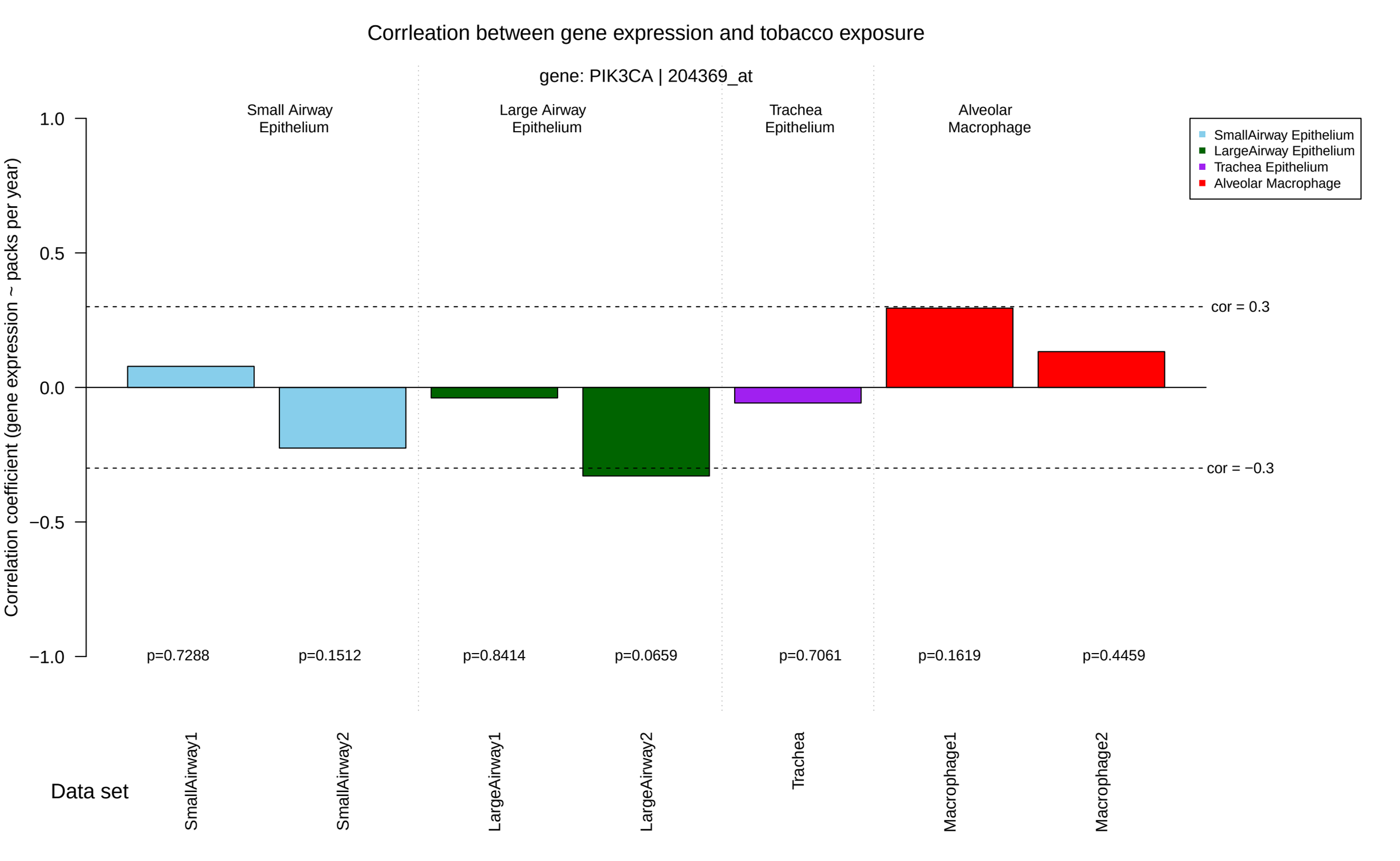

set: LargeAirway1 (p=0.0904)
PIK3CA | 204369_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0849)
PIK3CA | 204369_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2264)
PIK3CA | 204369_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1004)
PIK3CA | 204369_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2783)
PIK3CA | 204369_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIK3CA | 204369_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7288
p=0.1512
p=0.8414
p=0.0659
p=0.7061
p=0.1619
p=0.4459
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2